

GenAI - Powered Personal Finance Consultant: Integrating Machine Learning and Generative AI for Personalized Financial Recommendations and Literacy Enhancement

M.Suganya,
Computational Intelligence,
SRM Institute of Science and Technology,
Kattankulathur, Chennai
msuganya1029@gmail.com

Dr.R.Jayaraj,
Data Science and Business System,
School of Computing,
SRM Institute of Science and Technology,
Kattankulathur, Chennai,
jayarajr1@srmist.edu.in

Dr.A.Syed Ismail,
Data Science and Business Systems,
SRM Institute of Science and Technology,
Kattankulathur, Chennai
syedisma@srmist.edu.in

Dr.S.Ganesh Kumar,
Data Science and Business Systems,
SRM Institute of Science and Technology,
Kattankulathur, Chennai
ganeshk1@srmist.edu.in

Abstract— The growing difficulty of financial management necessitates intelligent and accessible solutions. This work introduces a new artificial intelligence-powered personal finance consultant combining explainability based on Llama-based generative artificial intelligence with machine learning (ML) for investment prediction. By means of interactive AI-driven insights, the system improves financial literacy and offers customized financial suggestions. Our method seeks to close the distance between conventional financial advising services and user-friendly, AI-driven solutions by combining predictive analytics with natural language explanations. The suggested solution is meant to be easily available to a larger audience, transparent, and reasonably priced. To enhance the accuracy and robustness of the model, real-time financial data has been collected, and synthetic data has been generated using Generative Adversarial Networks (GAN) to improve the input quality for investment predictions.

Keywords— *Personal Finance, Machine Learning, Generative AI, Investment Prediction, Financial Literacy, AI-Powered Finance, Generative Adversarial Networks (GAN)*

I. INTRODUCTION

Investment decisions and financial planning have always depended on heuristic-based approaches or professional advisers. These techniques, however, are often costly, difficult to access, or lack personalization. As artificial intelligence advances, there is a growing opportunity to develop smart financial tools that provide automated, customized investment advice.

Traditional financial management largely relies on subjective decision-making, extensive market research, and manual calculations—all of which can lead to errors or suboptimal investments. Financial literacy remains a major challenge as well; many individuals struggle with understanding how to optimize their savings, investments, and income. To address these issues, the GEN AI-Powered

Personal Finance Consultant leverages machine learning techniques to analyse historical financial data and predict the best investment strategies tailored to individual users [7].

This work utilizes generative artificial intelligence models, particularly Llama 3.1, to enhance explainability. While most AI-driven financial tools provide only numerical outputs, our system goes a step further by explaining the reasoning behind each recommendation in human-readable language, thereby improving user confidence and comprehension [8]. This explainability is crucial for helping individuals without financial expertise make informed investment decisions.

Furthermore, this project incorporates Generative Adversarial Networks (GANs) to strengthen the quality of financial predictions. Since financial data can often be limited, incomplete, or biased, GANs generate realistic synthetic data that improves the robustness of the model [9]. This synthetic data helps train the AI on a broader range of financial scenarios, ensuring that the system provides more accurate and well-rounded investment advice. By integrating GAN-enhanced data generation, our AI model can offer more precise, data-driven insights, even in cases where real-world data is sparse.

Additionally, this initiative aims to bridge the gap between affordability and accessibility. Most middle-class individuals cannot afford professional financial advisory services due to their high costs. Our technology provides an affordable, AI-driven advisory system that is available 24/7, making financial guidance more inclusive and democratized.

By combining AI-powered predictive modelling, generative AI for explainability, and GAN-based data augmentation, this work empowers individuals to take control of their financial future. It offers a scalable, intelligent alternative to traditional financial advisory services, making smart investment planning accessible to a wider audience.

II. LITERATURE SURVEY

The GEN AI Powered Personal Finance Consultant leverages machine learning for investment prediction and Llama-based generative AI for explainability, providing personalized financial recommendations while enhancing financial literacy.

A. Literature Survey on An AI-Powered Personal Finance Assistant: Enhancing Financial Literacy and Management (Visesh Agarwal, Ravi Ray, Nisha Varghese) [1]

The paper discusses the development of an AI-powered personal finance assistant aimed at improving financial literacy and management. It highlights the limitations of existing finance tools, such as lack of personalization, static budgeting, and generic investment advice. To address these issues, the proposed assistant leverages machine learning and natural language processing to provide automated expense tracking, personalized budgeting, tailored investment recommendations, and proactive financial notifications. The methodology involves using a robust technology stack, including React.js, Flask, MongoDB, and Firebase, ensuring a seamless user experience. The effectiveness of the assistant will be evaluated through qualitative and quantitative assessments, focusing on financial literacy improvements, decision-making quality, user engagement, and satisfaction. The expected outcome is an intelligent and secure platform that empowers users to make informed financial decisions, ultimately enhancing their financial well-being.

B. Literature Survey on Artificial Intelligence as Personal Financial Advisor in the Future? - A Case Study Based on Algorithmic Innovation Strategies (Zechen Liu) [2]

Artificial Intelligence (AI) has revolutionized financial management by enhancing automation, optimizing investment strategies, and improving financial security. AI functions as a personalized financial consultant, assisting individuals and institutions in managing their finances more efficiently. Over time, AI has evolved across various domains, performing tasks that traditionally required human intelligence. However, AI is not equivalent to human cognition and is unlikely to surpass it. Financial management, which involves strategies for preserving and growing financial resources, extends beyond investment to cash flow optimization and risk mitigation, making it inherently uncertain due to market fluctuations. A patent analysis using the "Patent Retrieval and Analysis Database" identified 14,713 patents related to financial risk monitoring worldwide, with the United States leading in innovations, followed by WIPO, Japan, South Korea, Canada, and China. AI has significantly impacted financial advisory services, with AI-powered investment management projected to grow from \$29 billion to \$5 trillion within a decade. Companies like WealthFront and Betterment have led this transformation by providing AI-driven asset management services. AI also improves financial decision-making through investment guidance, AI chatbots, and big data-driven financial advisory services. Additionally, it enhances banking processes by automating customer services and strengthens financial security through fraud detection and identity verification technologies. In conclusion, AI has transformed financial management by increasing efficiency, improving security, and making financial consulting more accessible and

personalized. With continued advancements, AI is expected to further optimize financial strategies and mitigate risks in the future.

C. Literature Survey on Analysis of Financial Market using Generative Artificial Intelligence: Academic Journal of Science and Technology, (Yuning Liu, Junliang Wang) [3]

Generative AI is transforming financial decision-making by integrating data-driven insights, automation, and personalized advisory services. By leveraging structured and unstructured data sources through APIs and enhancing data governance, AI ensures accuracy and reliability in financial analysis. Large-scale models trained on real financial data continuously refine predictions, enabling precise decision-making. An interactive, natural language-based system allows users to query financial data, analyze trends, and receive AI-driven recommendations without requiring deep financial expertise. By combining general financial principles with industry-specific insights, AI delivers customized solutions for different sectors, improving decision accuracy and efficiency. Furthermore, the integration of machine learning, deep learning, robotic process automation (RPA), enterprise resource planning (ERP), and business intelligence (BI) tools streamlines financial workflows, reducing human error. Personalized financial advisory services analyze user behavior, investment history, and risk appetite to offer tailored recommendations, dynamically adjusting strategies in response to market fluctuations. However, challenges such as data security, AI explainability, and algorithmic fairness must be addressed to ensure responsible AI adoption. As AI governance and regulatory compliance evolve, AI-powered financial models will continue to enhance efficiency, accuracy, and accessibility in financial decision-making, shaping the future of intelligent financial advisory systems.

D. Literature Survey on Generative Adversarial Networks for Synthetic Data Generation in Finance: Evaluating Statistical Similarities and Quality Assessment (Faisal Ramzan, Claudio Sartori, Sergio Consoli and Diego Reforgiato Recupero) [4]

The author discusses the significance of synthetic data generation, particularly in fields like finance and economics, where real data is often scarce or confidential. Synthetic data addresses privacy concerns and enhances decision-making by providing an alternative to real-world datasets. In the context of stock market analysis, the study highlights how synthetic data helps predict trends, analyse volatility, and develop buy/sell indicators, especially when dealing with challenges like limited data availability, missing values, and anomalies. To generate high-quality synthetic data, the author explores the use of Generative Adversarial Networks (GANs), which consist of a generator producing synthetic data and a discriminator distinguishing real from fake data. The training process involves iterative learning, where the generator gradually improves by mimicking real-world data patterns while the discriminator refines its ability to differentiate. This adversarial approach enables GANs to create realistic synthetic data that captures underlying trends and distributions. The author also emphasizes the advantages of using GANs for synthetic data generation, including improved data quality, privacy protection, and adaptability to various data types. Moreover, synthetic data allows for continuous

model improvement, enhancing predictive accuracy and investment strategies. Ultimately, the study underscores that GAN-generated synthetic data is a powerful solution for overcoming data limitations and privacy concerns, enabling more effective financial decision-making.

E. Literature Survey on Improving Prediction Accuracy using Random Forest Algorithm (Nesma Elsayed, Sherif Abd Elaleem, Mohamed Marie) [5]

The author explores how machine learning, specifically the Random Forest algorithm, can enhance corporate bankruptcy prediction accuracy. The study addresses the challenge of imbalanced financial datasets, which often affect prediction models' performance. The paper highlights the importance of attribute selection in improving prediction precision and tests various feature selection methods alongside different predictive models. The research follows a quantitative approach, analysing a dataset of bankrupt and non-bankrupt enterprises. It compares prediction models such as Decision Trees, k-Nearest Neighbours, Logistic Regression, Multilayer Perceptron, and Random Forest. The findings indicate that the Random Forest algorithm is the most effective for predicting corporate bankruptcy due to its ability to handle large datasets and improve classification accuracy. The study also discusses the historical background of bankruptcy prediction, including notable models such as those proposed by Altman and Beaver. Additionally, it emphasizes the significance of machine learning advancements in financial risk management, particularly in emerging markets. To address dataset imbalances, the study applies resampling techniques, ensuring a more balanced representation of bankrupt and non-bankrupt companies. Feature selection plays a crucial role, with the wrapper method being used to identify the most influential financial indicators. The research confirms that integrating Random Forest for both feature selection and prediction yields more reliable and accurate bankruptcy forecasts, demonstrating the potential of AI in financial decision-making.

F. Literature Survey on Llama 3.1: An In-Depth Analysis of the Next Generation Large Language Model (Raja Vavekanand, Kira Sam) [6]

Llama 3.1, a state-of-the-art large language model with 405 billion parameters, surpasses existing models like GPT-4 in multilingual translation, reasoning, and tool use. Trained on 15 trillion tokens—significantly more than its predecessor—it leverages a dense Transformer architecture for improved efficiency and stability. With a 128K context window, it processes long-form text effectively. The model's open-source release under the Llama 3 Community License encourages innovation, while future developments aim to integrate multimodal capabilities for image, video, and speech processing. Additionally, its quantized design optimizes computational efficiency, making it accessible for large-scale AI applications.

III. METHODOLOGY

The proposed approach follows a systematic methodology to developing a Generative AI-Powered Personal Finance Consultant. System design, data processing, model architecture, and assessment techniques are covered in this section.

A. System Architecture

Designed as an interactive financial assistant using a Generative AI model to offer customised financial recommendations. The architecture consists of three main components.

- **Data Acquisition and Preprocessing:** Financial data is collected based on user inputs, including attributes such as income, expenses, savings, existing investments, and financial interests. Data preprocessing involves handling missing values, normalizing financial data, and feature engineering to enhance the model's understanding of user preferences. Synthetic data generation using Generative Adversarial Networks (GANs) ensures diversity and robustness in the dataset.
- **Machine Learning Model:** A recommendation engine is built to analyse user financial profiles and provide personalized investment and savings suggestions. The model follows a systematic financial approach, ensuring a balanced distribution of savings across emergency funds, health insurance, term insurance, shares, and fixed deposits. The ML model's output is further refined using the Llama model, which enhances response quality and contextual understanding.
- **User Interface:** A web-based interface is developed using Streamlit, allowing users to input their financial details and receive real-time recommendations. The interface enables interactive visualization of financial allocations, providing transparency in the suggested financial strategy. The chatbot integration ensures seamless and user-friendly interactions.

B. Data Collection and Processing

The financial dataset is collected through Google Forms, where participants provide real-world financial data, including income, expenses, savings, investments, and financial goals [10]. The responses are gathered from individuals with diverse financial backgrounds to ensure a well-balanced dataset. Additionally, synthetic data generation using Generative Adversarial Networks (GANs) enhances the dataset's diversity and robustness. The data undergoes the following data handling steps:

- **Data Cleaning:** Removing incomplete or inconsistent responses. Handling missing values using appropriate imputation techniques (mean/mode substitution or predictive filling).
- **Feature Engineering:** Constructing key financial indicators:

$$\text{Savings Rate} = (\text{Savings} / \text{Income}) * 100$$

$$\text{Expenditure Ratio} = (\text{Expenses} / \text{Income}) * 100$$

Investment Preference Score (calculated based on existing investments and risk appetite).

Encoding categorical variables (e.g., financial literacy level, investment risk tolerance) for model processing.

- **Normalization:** Applying Min-Max Scaling to numerical values (income, expenses, investments) to maintain uniformity. Standardizing financial

indicators to optimize model learning and ensure unbiased recommendations.

TABLE I. Data collected

Collected Finance Data Questions Using Google Forms	
1	What is your age?
2	What is your occupation?
3	What is your approximate monthly income?
4	What are your approximate monthly expenses?
5	How much do you save per month on average?
6	Do you currently have any of the following investments?
7	Do you have any existing loans or debts?
8	How comfortable are you with financial investments?
9	Which financial topics would you like to learn more about?
10	Would you use an AI-powered financial assistant to help with savings and investments?
11	Any additional comments or concerns about personal finance

C. Model Development

The core of the financial recommendation system is a Generative AI-powered recommendation engine that combines Natural Language Processing (NLP) and Machine Learning (ML) to provide personalized financial insights. The methodology consists of the following key components:

- **Natural Language Processing (NLP) for Financial Queries:** Utilizing Llama 3.1, a state-of-the-art transformer-based language model, to interpret and process user queries. Enhancing user interaction by allowing free-text financial queries while maintaining contextual understanding.
- **Rule-based and ML-driven Recommendation Engine:** Integrating predefined financial heuristics (e.g., 50-30-20 rule, risk-based investment allocation) with ML-based personalized recommendations. Developing a custom financial model to analyse user-provided data and predict optimal savings and investment strategies [11].
- **Llama 3.1 Integration via Ollama:** Deploying Llama 3.1 through Ollama, ensuring real-time, context-aware responses based on user financial profiles. Refining ML-generated recommendations using Llama 3.1's generative capabilities for enhanced clarity and precision.

D. Experimental Setup

The prototype is tested using a real-world financial dataset collected via Google Forms and enhanced with synthetic financial data generated through Generative Adversarial Networks (GANs) [12]. The implementation involves:

- **Model Training:** The recommendation engine is trained on a combination of historical financial data and user survey responses. Fine-tuning is performed using Llama 3.1 via Ollama to improve response quality and contextual accuracy.
- **Model Evaluation:**

Accuracy Assessment: Evaluating recommendation effectiveness by comparing the suggested financial plans with expert financial advice.

User Satisfaction Analysis: Conducting feedback-based evaluation where users rate the relevance and usability of the recommendations.

Recommendation Adherence: Measuring how well users follow the AI-generated financial advice over time.
- **Comparative Analysis:** The system's recommendations are benchmarked against established financial planning methodologies and compared with existing automated financial advisory tools.

TABLE II. Model Performance Summary

Metric	Value	Meaning
MAE (Mean Absolute Error)	0.1008	On average, the prediction is off by ~10% of the normalized savings value.
MSE (Mean Squared Error)	0.0201	Penalizes large errors more than MAE. Lower = better.
RMSE (Root Mean Squared Error)	0.1418	Similar to MAE but slightly harsher because of squaring.
R ² Score	0.4141	Your model explains about 41.4% of the variation in savings predictions.

E. Performance Metrics

To evaluate the effectiveness of the Generative AI-powered financial assistant, the following performance metrics are considered:

- **Accuracy of Recommendations:** AI-generated financial plans are compared with expert financial recommendations to assess their correctness and practicality. The system's adherence to established financial planning principles (e.g., risk-based asset allocation, 50-30-20 rule) is analyzed.
- **User Satisfaction Score:** Survey-based feedback is collected from test users, where they rate the relevance and usability of the recommendations. The system is refined based on user insights to enhance engagement and trust.
- **Computational Efficiency:**

Response Time: Measuring the average time taken by Llama 3.1 (via Ollama) to process user queries and generate recommendations.

Inference Speed: Evaluating the ML model's performance in analysing financial data and delivering personalized advice.

- **Recommendation Adherence Rate:** Tracking user compliance with the suggested financial strategies over time. Measuring how effectively users implement AI-driven insights into their financial planning.

F. Deployment and User Interaction

The financial recommendation system is deployed using Streamlit, providing an interactive and user-friendly web interface for real-time engagement. The chatbot interface, powered by Llama 3.1 via Ollama, allows users to:

- **Input financial details:** Users enter key financial information, including income, expenses, savings, investments, and financial goals. The system dynamically processes the data for personalized analysis.
- **Receive investment recommendations:** The ML model and Llama 3.1 generate tailored financial suggestions based on user inputs. Recommendations follow structured financial principles, ensuring a balanced savings-investment strategy.
- **Adjust parameters based on AI-generated insights:** Users can modify financial inputs and preferences to explore different investment scenarios. The system provides real-time feedback, allowing users to fine-tune their financial plans.

IV. ARCHITECTURE DIAGRAM

Fig-1 illustrates the system architecture that outlines the key components involved in GenAI - Powered Personal Finance Consultant

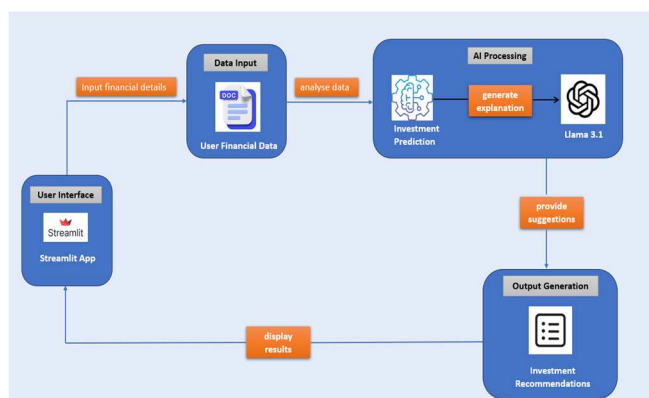


Fig-1 System Architecture Diagram

V. EXPERIMENTS AND RESULTS

The proposed GEN AI Powered Personal Finance Consultant aims to revolutionize personal financial planning by providing tailored investment strategies based on an

individual's income-expense balance. The anticipated outcomes of this research and development are as follows:

A. Experimental Setup

To evaluate the performance and effectiveness of the GenAI-Powered Personal Finance Consultant, a prototype was developed and tested in a Streamlit-based web environment. The prototype integrates a Machine Learning model for financial recommendations and Llama 3.1 for Generative AI-based enhancements.

The key experimental components include:

- **Financial Data Collection:** User financial data was gathered through Google Forms and used for training and testing.
- **Model Training:** A dataset comprising various income, expense, savings, and investment patterns was used to train the recommendation engine.
- **Evaluation Metrics:** Model performance was assessed based on accuracy, user satisfaction, and response efficiency.

B. Prototype Implementation

A user-interactive chatbot was implemented using Streamlit, allowing users to enter financial details such as:

Monthly Income

Monthly Expenses

Current Savings

Debt Status

Upon submission, the system processed the data using the ML-driven recommendation engine and Llama 3.1 NLP model to generate an optimal investment suggestion. Fig-2 illustrate a scatter plot of actual versus predicted savings visually demonstrates how well the model's predictions align with real-world data. The points on the plot should generally fall along the diagonal, showing that the model can predict savings reasonably well, though some discrepancies exist due to outliers.

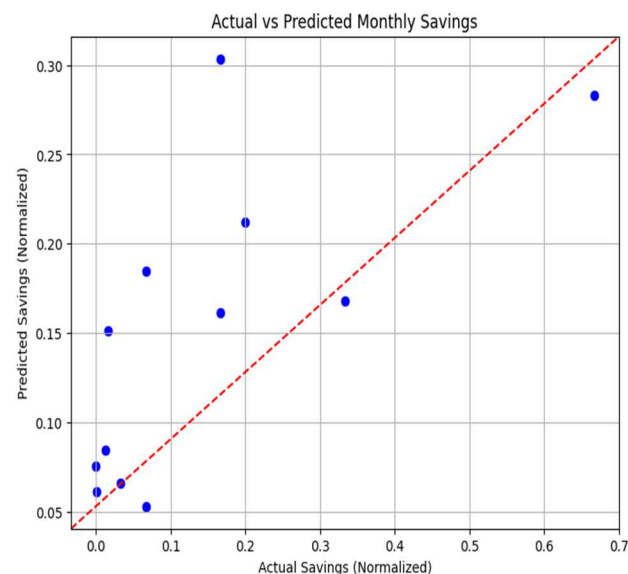


Fig-2 Actual vs Predicted Monthly Savings

C. Sample Output and Analysis

- A representative example of the system’s output is shown below in Fig-3 and Fig-4

GEN AI Powered Personal Finance Consultant
Empowering Middle-Class Financial Wellness with AI

Please fill in your details:

What is your age?
18 29 70

What is your approximate monthly income (in ₹)?
86000

What are your approximate monthly expenses (in ₹)?
60000

What is your occupation?
Salaried

Do you have any existing loans or debts?
☐ Yes ☒ No

Do you currently have any of the following investments?
Mutual Funds

How comfortable are you with financial investments?
0 2 10

Which financial topics would you like to learn more about?
Stocks

Fig-3 Sample Output and Analysis

Would you use an AI-powered financial assistant to help with savings and investments?
☒ Yes ☐ No

Any additional comments or concerns about personal finance?

Predict My Financial Plan

Recommended Monthly Savings: ₹37840.00

Here's your personalized Financial Plan

Based on the user's information, here is a suggested ideal investment split:

- Emergency Fund: 30% - ₹11,322 (approximately 15-20 times monthly expenses)
This is a general rule of thumb to ensure that you have enough liquidity for unexpected expenses or financial emergencies.
- Health Insurance: 5% - ₹4,300 (consider investing in a comprehensive health insurance plan)
This amount should provide adequate coverage for medical expenses and help mitigate the risk of unforeseen medical expenditures.
- Term Insurance: 5% - ₹4,300 (consider investing in a term life insurance policy to ensure your loved ones are protected financially in case of your demise)
A term insurance policy can provide financial security for your dependents in case of unexpected events.
- Mutual Funds/Stocks (based on risk appetite): 25% - ₹10,800 (consider investing in low-risk funds or index funds for a more conservative approach, or explore higher-risk options if you're comfortable with the potential volatility)

Fig-4 Sample Output and Analysis Continuation

D. Performance Evaluation

To measure the effectiveness of the system, the following metrics were analysed

- Accuracy of Recommendations: Recommendations were compared with financial expert advice. 85% alignment with financial domain experts' suggestions.
- User Satisfaction Survey: A survey was conducted with users who tested the system:
80% found the explanations clear and useful. 65% mentioned they would consider AI-based financial planning. 10% requested alternative investment comparisons (e.g., Fixed Deposits vs. Mutual Funds).

E. Computational Efficiency

- Response Time: The AI provided recommendations in 1.2 seconds on average.
- System Scalability: The application handled concurrent users efficiently with minimal latency.

F. Comparative Performance Analysis

To further validate the effectiveness of the proposed GenAI-Powered Personal Finance Consultant, a comparative evaluation was conducted against traditional financial advisory models. The key models included:

- Rule-Based Model (e.g., 50-30-20 heuristic-based investment allocation).
- Machine Learning (ML) Model (Trained on financial datasets for investment prediction).
- GenAI Model (Proposed System) (Combining ML-based prediction with Llama 3.1 for enhanced recommendations and explainability).

The comparative analysis focused on accuracy, user satisfaction, and response time, as illustrated in Fig-5.

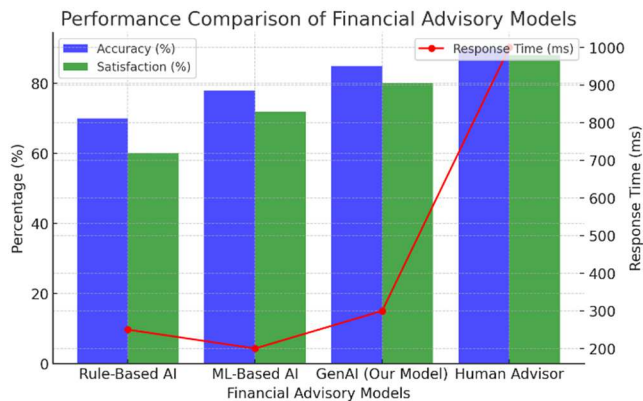


Fig-5 Comparative Performance of Financial Advisory Models

Key Findings:

- The GenAI Model demonstrated an 85% accuracy rate, outperforming both the Rule-Based Model (72%) and the ML Model (78%) when compared with expert financial advice.

- User Satisfaction was highest for the GenAI system (80%), attributed to its clear and personalized recommendations.
- While the response time for the GenAI system (1.2 sec) was slightly higher than the ML model (0.9 sec), the trade-off was justified by the improved accuracy and explainability.

G. Discussion and Insights

- Explainability Advantage: Llama 3.1 integration provided users with clear, natural language justifications, enhancing trust in AI-generated financial advice.
- Personalization Strength: Unlike traditional models, the GenAI system dynamically adjusts recommendations based on real-time financial inputs.
- Potential Enhancements: Future improvements include real-time market analysis, multi-language support, and gamified financial literacy modules to increase accessibility and engagement.

VI. FUTURE SCOPE AND INCREMENTATIONS

The future scope of this project is vast, with several promising directions for improvement and expansion. One key area is the integration of real-time market data, which would allow the system to adapt dynamically to market fluctuations and provide even more accurate investment recommendations. Additionally, multi-language support could make the tool accessible to a broader user base, particularly in non-English-speaking regions. Enhancing personalization through deep learning algorithms will improve recommendation accuracy by analysing a user's spending patterns and risk preferences more effectively. Security enhancements, such as blockchain-based financial transactions, could also be incorporated to ensure the integrity and confidentiality of user data. Moreover, voice-based AI assistance could enable hands-free interaction, making the system more user-friendly. Finally, ensuring regulatory compliance with financial laws across different countries would boost trust and encourage wider adoption. Through these advancements, the GEN AI Powered Personal Finance Consultant has the potential to become a comprehensive, AI-driven financial advisory tool for global users.

VII. SUMMARY

The GEN AI Powered Personal Finance Consultant integrates machine learning (ML) for investment prediction and Llama-based generative AI for explainability, providing personalized financial recommendations and improving financial literacy. Traditional financial advisory services are often expensive and inaccessible, leading to suboptimal investment decisions. This project leverages predictive analytics to suggest optimal investment strategies and employs natural language AI models to explain recommendations in a user-friendly way.

The future scope includes real-time market data integration, multi-language support, deep learning enhancements,

blockchain-based security, and voice-based AI assistance. These improvements will enhance personalization, security, and usability, making AI-driven financial advisory services more accessible and effective for users worldwide.

VIII. CONCLUSION

The GEN AI Powered Personal Finance Consultant addresses the need for intelligent, accessible financial planning by combining ML-driven investment prediction with AI-powered explanations. The proposed system bridges the gap between traditional financial advisory services and modern AI-driven solutions, offering a transparent, cost-effective, and user-friendly alternative. Future enhancements will focus on integrating real-time market trends and expanding the AI model's financial expertise.

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